

*Arrival Price:* The arrival price algorithm has different meanings across different brokers and vendors. So it is important to speak with those parties to understand the exact specifications of these algorithms. But in general, the arrival price algorithm is a cost minimization strategy that is determined from an optimization that balances the trade-off between cost and risk (e.g., Almgren and Chriss, 1997). Users specify their level of risk aversion or trading urgency. The resulting solution to the optimization is known as the trade schedule or trade trajectory and is usually front-loaded. However, some parties solve this optimization based on a POV rate rather than a static schedule in order to take advantage of changing liquidity patterns.

*Implementation Shortfall:* The implementation shortfall algorithm is similar to the arrival price algorithm in many ways. First, its meaning varies across the different brokers and different vendors and so it is important to speak with those parties to understand their exact specifications. Second, we base the implementation shortfall algorithm on Perold's (1988) paper and seek to minimize cost through an optimization that balances the trade-off between cost and risk at a user specified level of risk aversion. In the early days of algorithms trading, the arrival price and implementation shortfall algorithms were identical across different brokers. Thus, to distinguish implementation shortfall from arrival price, brokers began to incorporate real-time adaptation tactics into the implementation shortfall logic. These rules specify how the initial solution will deviate from the optimally prescribed strategy in times of changing market liquidity patterns and market prices. Thus, while arrival price and implementation shortfall still do not have a standard definition across the industry, the general consensus is that the arrival price algorithm is constant while the implementation shortfall algorithm incorporates a second level of adaptation tactics based on market volumes and market prices.

*Basket Algorithms:* Basket algorithms, also known as portfolio algorithms, are algorithms that manage the trade-off between cost and total basket risk based on a user specified level of risk aversion. These algorithms will manage risk throughout the trading day and adapt to the changing market conditions based on user specifics. The algorithms are usually based on a multi-trade period optimization process. They make real-time trading decisions based on how those decisions will affect the overall performance of the basket. For example, a basket algorithm may choose to not accelerate trading in an order even when faced with available liquidity and favorable prices if doing so would increase the residual risk of the basket. Furthermore, the basket algorithm may be more